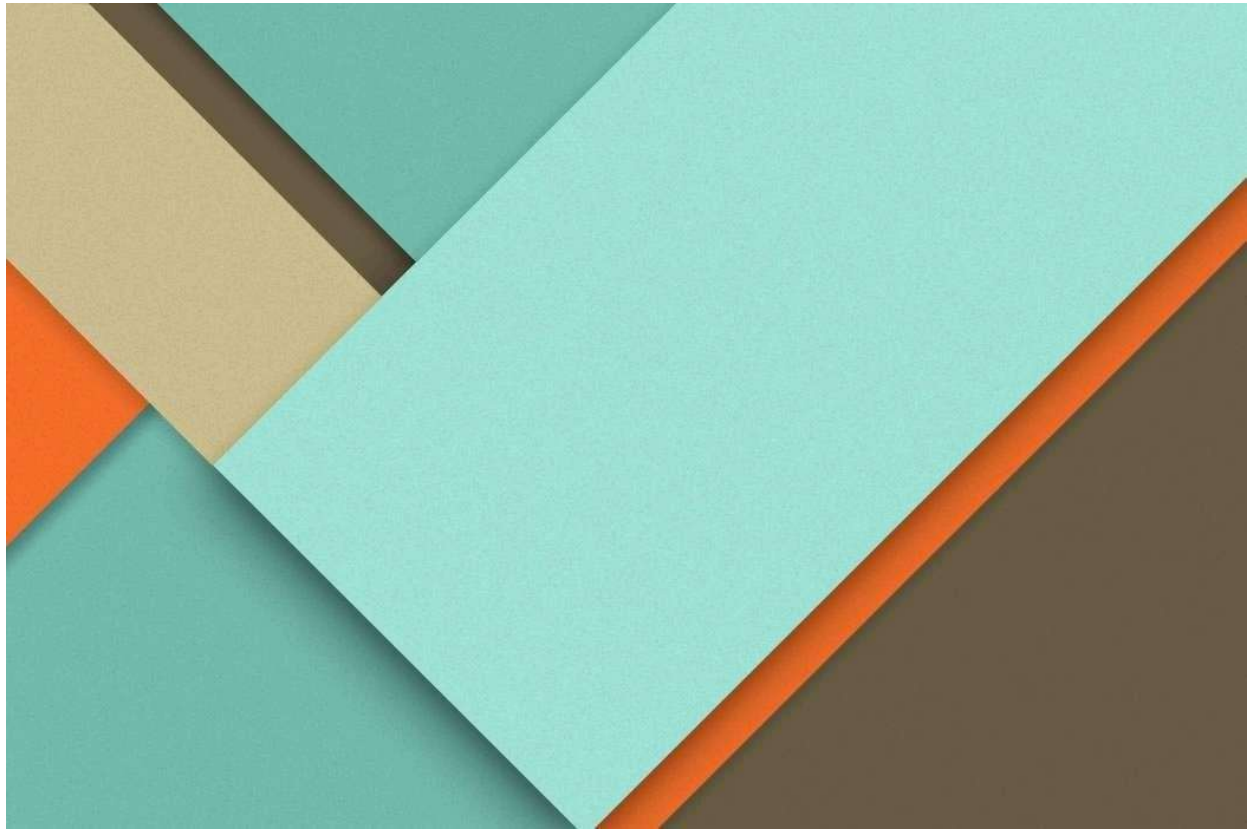




CLOUD COMPUTING

LAB-7

17-01-2025

GITHUB: <https://github.com/akshayag2005/CLOUD-COMPUTING>

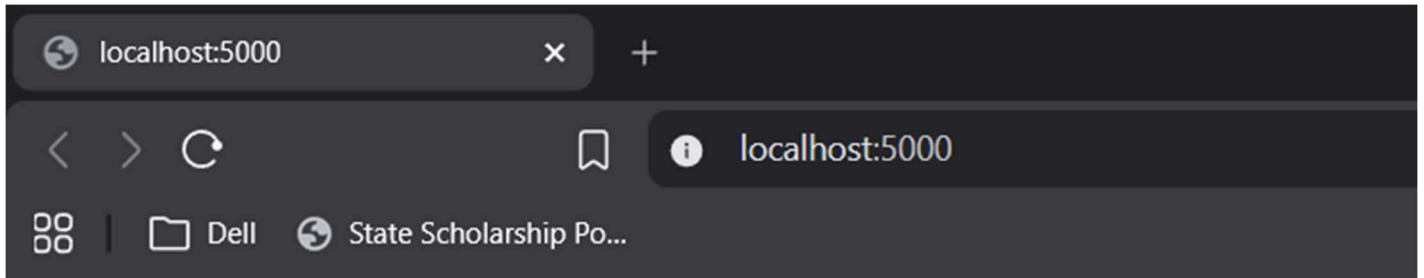
AKSHAY

PES2UG23CS044

CSE-A

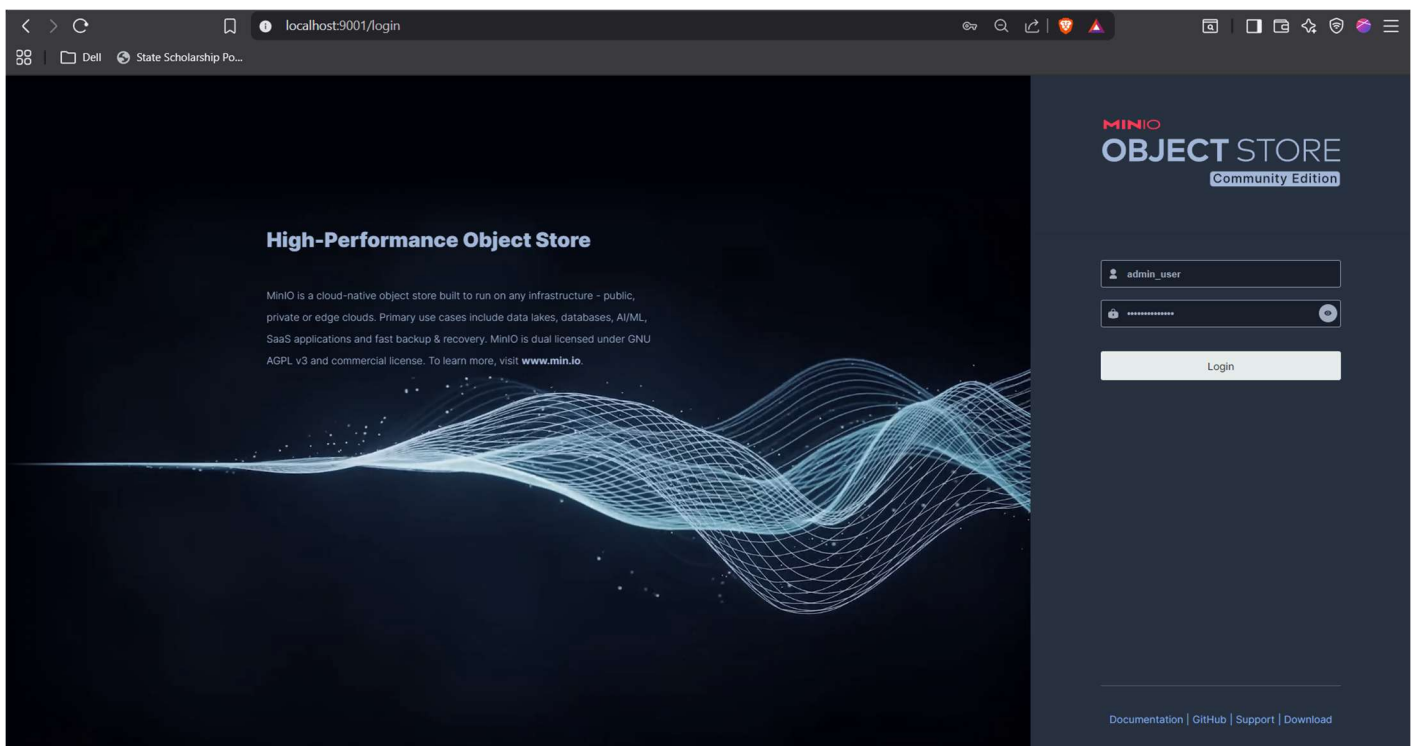
PART 1: Setup

SS-1



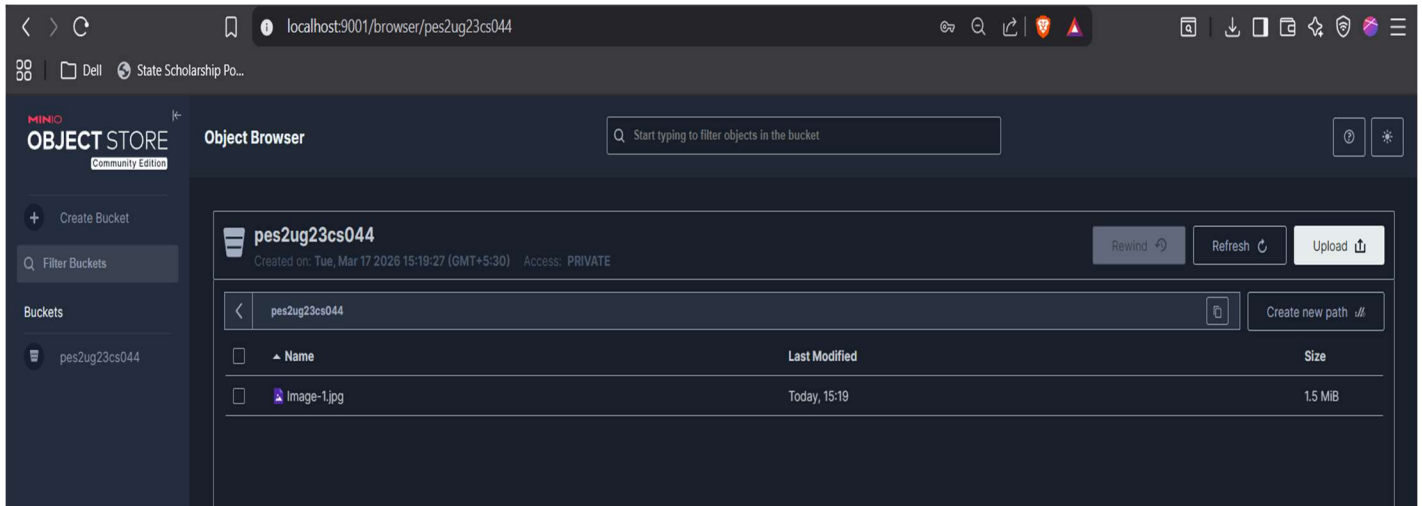
E-commerce Lab is Running! Send a POST request to /product to add data.

SS-2



PART 2: Upload Product

SS-3



PART 3: EXPERIMENTS

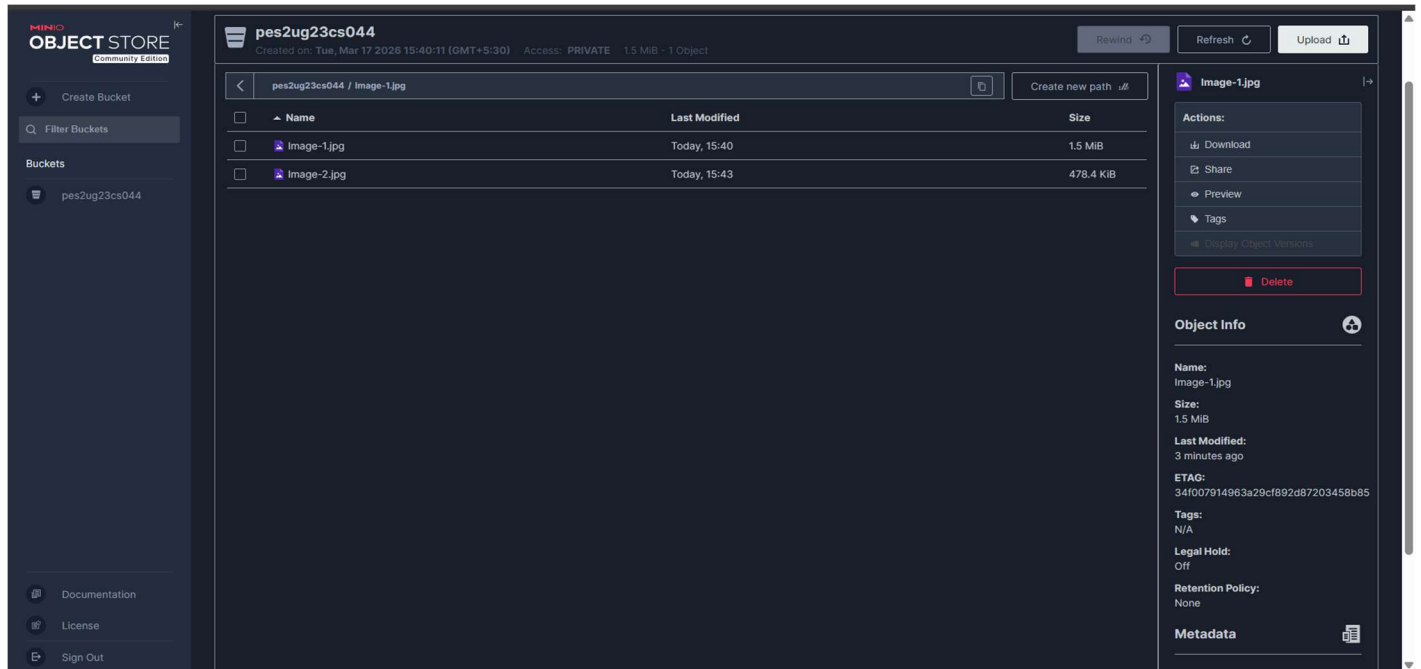
Experiment 1: Access Method

SS-4

```
AKSHAY_AG@AKSHAY: D:\SEM 6\Cloud computing\CC Lab\CC LAB 7\PES2UG23CS044$ docker exec -it pes2ug23cs044-webapp-1 python -c "import sqlite3; conn = sqlite3.connect('/mnt/block_volume/e-commerce.db'); cursor = conn.cursor(); cursor.execute('SELECT * FROM product'); print(cursor.fetchall()); conn.close()"
[(1, 'Phone', 500.0, 'Image-1.jpg')]
```

```
AKSHAY_AG@AKSHAY: D:\SEM 6\Cloud computing\CC Lab\CC LAB 7\PES2UG23CS044$
```

Experiment 2: Metadata SS-5



Experiment 3: Scaling & Fragmentation

Observation:

- When the block storage (local volume) is filled, the application throws a database full or disk error.
- The database cannot accept new entries once the storage limit is reached.
- MinIO continues to accept image uploads as long as the host system has available storage.

Conclusion:

- Block storage has fixed capacity limitations.
- Object storage is scalable and can expand as needed.

Experiment 4: Recovery & High Availability

SS-7

```
AKSHAY_AG@AKSHAY D:\SEM 6\Cloude computing\CC Lab\CC LAB 7\PES2UG23CS044 docker-compose stop minio
time="2026-03-17T15:48:39+05:30" level=warning msg="D:\\SEM 6\\Cloude computing\\CC Lab\\CC LAB 7\\PES2UG23CS044\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] stop 1/1
✓ Container pes2ug23cs044-minio-1 Stopped 0.4s
AKSHAY_AG@AKSHAY D:\SEM 6\Cloude computing\CC Lab\CC LAB 7\PES2UG23CS044 curl -X POST -F "name=Phone" -F "price=500" -F "image=@Image-2.jpg" http://localhost:5000/product
<!doctype html>
<html lang=en>
<title>500 Internal Server Error</title>
<h1>Internal Server Error</h1>
<p>The server encountered an internal error and was unable to complete your request. Either the server is overloaded or there is an error in the application.</p>
AKSHAY_AG@AKSHAY D:\SEM 6\Cloude computing\CC Lab\CC LAB 7\PES2UG23CS044 |
```



LAB OBSERVATIONS

Experiment 1: Identifying the Access Method

Observation:

- The database file ecommerce.db is visible in the local folder (my_block_data).
- The uploaded image file is not present in the local file system.
- The image can only be accessed through the MinIO dashboard.
- Querying the database shows structured data like product name, price, and image name, but not the actual image content.

Experiment 2: Metadata vs File Headers

Observation:

- After adding metadata, the uploaded image in MinIO shows additional fields such as:
 - product-name
 - product-price
- These metadata values are visible directly in the MinIO UI without opening the file.
- The database (.db file) does not store or display such metadata directly.



Experiment 3: Scaling & Fragmentation

Observation:

- When the block storage (local volume) is filled, the application throws a database full or disk error.
- The database cannot accept new entries once the storage limit is reached.
- MinIO continues to accept image uploads as long as the host system has available storage.

Experiment 4: Recovery & High Availability

Observation:

- When MinIO service is stopped, image uploads fail, but database operations still work.
- When the web application is stopped, the data stored in the database remains intact.
- Restarting services restores functionality without data loss.

The experiment demonstrates the fundamental differences between block and object storage systems. Block storage provides structured, locally accessible data suitable for databases, while object storage offers scalable, metadata-rich storage for unstructured data such as images. The separation of storage and application layers ensures better scalability, reliability, and flexibility in modern cloud architectures.